MAT1575, Classwork12, Fall2026

6.3 Taylor and Maclaurin Polynomials (p.497-512)

1. *Review*: Sequences and Series:

A Sequence is an enumerated list of numbers and it can be denoted by $a_1, a_2, a_3, \cdots$, or $\{a_n\}$ or $\{a_n\}_{n\geq 1}^{\infty}$

Let $\{a_k\}$ be a sequence. The series is the **sum** of all the term of a_k for $k \geq 1$: $\sum_{k=1}^{\infty} a_k$

Example: $\{1, 2, 100, 3, 4, 1000, \cdots\}$ or $\{1, 4, 9, 16, 25, 36, 49, 64, \cdots\}$

2. *Definition*: Power Series.

A series of the form is called a power series centered at $x=0$ if the form looks like: $\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \cdots$

A series of the form is called a power series centered at $x=a$ if the form looks like:

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + c_3 (x-a)^3 + c_4 (x-a)^4 + \cdots$$

3. A Power Series Representation of a Function.

If a function $f(x)$ is good enough (?), then this function $f(x)$ has

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + c_3 (x-a)^3 + c_4 (x-a)^4 + \cdots$$

Why do we need this? Divide and Conquer: Knowing Polynomials = knowing the approximation of $f(x)$

4. How to find out the coefficients $c_0, c_1, c_2, c_3, c_4, c_5, \ldots$ in the power series representation?

For c_0, since $f(x) = c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + c_4(x-a)^4 + c_5(x-a)^5 + c_6(x-a)^6 + \cdots$

then when $x = a$, we have $f(a) = c_0 + 0 + 0 + 0 + \cdots$ which implies $c_0 = f(a)$

For c_1, since $\frac{d}{dx} f(x) = c_1 + 2 \cdot c_2 (x-a) + 3 c_3 (x-a)^2 + 4 \cdot c_4 (x-a)^3 + 5 \cdot c_5 (x-a)^4 + 6 \cdot c_6 (x-a)^5 + \cdots$

then when $x=a$, we have $\frac{d}{dx} f(x)\Big|_{x=a} = f'(a) = c_1$ which implies $c_1 = f'(a)$

For c_2, since $\frac{d^2}{dx^2} f(x) = \frac{d}{dx}\left(\frac{d}{dx} f(x)\right) = 2 \cdot 1 \cdot c_2 + 3 \cdot 2 \cdot c_3 (x-a) + 4 \cdot 3 \cdot c_4 (x-a)^2 + 5 \cdot 4 \cdot c_5 (x-a)^3 + 6 \cdot 5 \cdot c_6 (x-a)^4 + \cdots$

then when $x=a$, we have $\frac{d^2}{dx^2} f(x)\Big|_{x=a} = f''(a) = 2 \cdot 1 \cdot c_2$ which implies $c_2 = \frac{f''(a)}{2 \cdot 1} = \frac{f''(a)}{2!}$

For c_3, since $\frac{d^3}{dx^3} f(x) = \frac{d}{dx}\left(\frac{d^2}{dx^2} f(x)\right) = 3 \cdot 2 \cdot 1 \cdot c_3 + 4 \cdot 3 \cdot 2 \cdot c_4 (x-a) + 5 \cdot 4 \cdot 3 \cdot c_5 (x-a)^2 + 6 \cdot 5 \cdot 4 \cdot c_6 (x-a)^3 + \cdots$

then when $x=a$, we have $\frac{d^3}{dx^3} f(x)\Big|_{x=a} = f'''(a) = 3 \cdot 2 \cdot 1 \cdot c_3$ which implies $c_3 = \frac{f'''(a)}{3 \cdot 2 \cdot 1} = \frac{f'''(a)}{3!}$

For c_4, since $\frac{d^4}{dx^4} f(x) = \frac{d}{dx}\left(\frac{d^3}{dx^3} f(x)\right) = 4 \cdot 3 \cdot 2 \cdot 1 \cdot c_4 + 5 \cdot 4 \cdot 3 \cdot 2 \cdot c_5 (x-a) + 6 \cdot 5 \cdot 4 \cdot 3 \cdot c_6 (x-a)^2 + \cdots$

then when $x=a$, we have $\frac{d^4}{dx^4} f(x)\Big|_{x=a} = f^{(4)}(a) = 4 \cdot 3 \cdot 2 \cdot 1 \cdot c_4$ which implies $c_4 = \frac{f^{(4)}(a)}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{f^{(4)}(a)}{4!}$

For c_n, we have $c_n = \frac{f^{(n)}(a)}{n!}$

Therefore,

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \frac{f^{(4)}(a)}{4!}(x-a)^4 + \frac{f^{(5)}(a)}{5!}(x-a)^5 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$$

5. *Definition*: Taylor Series.

If a function f has derivatives of all order at $x=a$, then the Taylor series for f at $x=a$ is

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n = f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f'''(a)}{3!}(x-a)^3+\frac{f^{(4)}(a)}{4!}(x-a)^4+\cdots$$

The Taylor Series for f at $x=0$ is known as Maclaurin Series for f

$$f(x)=\sum_{n=0}^{\infty}\frac{f^{(n)}(0)}{n!}x^n = f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4+\cdots$$

6. Examples. Find the Maclaurin Series for each given function. power series at $x=0$

(a) $f(x)=\sin(x)$ $c_0=f(0)=0$ $f'(x)=\cos(x)$; $f'(0)=1$; $f''(x)=-\sin(x)$ $f''(0)=0$; $f'''(x)=-\cos(x)$ $f'''(0)=-1$

$$\sin(x)=f(0)+f'(0)(x-0)+\frac{f''(0)}{2!}(x-0)^2+\frac{f'''(0)}{3!}(x-0)^3+\frac{f^{(4)}(0)}{4!}(x-0)^4+\frac{f^{(5)}(0)}{5!}(x-0)^5$$
$$=0+x+0+\left(\frac{-1}{3!}\right)x^3+0+\cdots$$
$$\sin(x)=x-\frac{1}{3!}x^3+\frac{1}{5!}x^5+\cdots$$

(b) $f(x)=\cos(x)$ $c_0=f(0)=1$. $f'(x)=-\sin(x)$. $f'(0)=0$ $f''(x)=-\cos(x)$, $f'''(x)=\sin(x)$, $f'''(0)=0$; $f^{(4)}(x)=\cos(x)$ $f''(0)=-1$ $f^{(4)}(0)=1\cdots$

$$\cos(x)=1+\frac{0}{1!}x+\frac{-1}{2!}x^2+\frac{0}{3!}x^3+\frac{1}{4!}x^4+\frac{0}{5!}x^5-\frac{1}{6!}x^6+\cdots=1-\frac{1}{2!}x^2+\frac{x^4}{4!}-\frac{x^6}{6!}+\cdots$$

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(b) $f(x)=e^x$, $f'(x)=e^x$, $f''(x)=e^x$, $f'''(x)=e^x$, $f^{(4)}(x)=e^x$,

$f(0)=e^0=1$ $f'(0)=e^0=1$ $f''(0)=e^0=1$ $\cdots$

$$f(x)=1+1\cdot x+\frac{1}{2!}x^2+\frac{1}{3!}x^3+\frac{1}{4!}x^4+\frac{1}{5!}x^5+\cdots$$

7. *Definition*: Taylor Polynomials. Given a function f which has derivatives of all orders at $x=a$.

Then the nth partial sum of the Taylor series for f at $x=a$ is known as Taylor Polynomials or Taylor polynomial of degree n:

$$P_n(x)=f(a)+\frac{f'(a)}{1!}(x-a)+\frac{f''(a)}{2!}(x-a)^2+\cdots+\frac{f^{(n)}(a)}{n!}(x-a)^n$$

The Taylor Polynomial for f at $x=0$ is known as the nth Maclaurin polynomial or Maclaurin polynomial of degree n:

$$P_n(x)=f(0)+\frac{f'(0)}{1!}x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\cdots+\frac{f^{(n)}(0)}{n!}x^n$$

8. Example. Find the Taylor Polynomial of degree 3 for $f(x)=\sin x$ at $a=\frac{\pi}{2}$. $f'(x)=\cos(x)$, $f''(x)=-\sin(x)$ $f'''(x)=-\cos(x)$

$P_0=f(\frac{\pi}{2})=\sin(\frac{\pi}{2})=1$

$P_1=f(\frac{\pi}{2})+f'(\frac{\pi}{2})(x-\frac{\pi}{2})=f(\frac{\pi}{2})+0=1$ ($f'(\frac{\pi}{2})=0$)

$$P_3=f(\tfrac{\pi}{2})+\frac{f'(\frac{\pi}{2})}{1!}(x-\tfrac{\pi}{2})+\frac{f''(\frac{\pi}{2})}{2!}(x-\tfrac{\pi}{2})^2+\frac{f'''(\frac{\pi}{2})}{3!}(x-\tfrac{\pi}{2})^3=1-\frac{1}{2!}(x-\tfrac{\pi}{2})^2$$

($f'(\frac{\pi}{2})=0$, $f''(\frac{\pi}{2})=-1$, $f'''(\frac{\pi}{2})=0$)

9. Taylor's Theorem with Remainder.

Let f be a function that can be differentiated $n+1$ times on an interval I containing the real number a. Let p_n be the nth Taylor polynomial of f at a and let

$$R_n(x)=f(x)-p_n(x)$$

be the nth remainder. Then for each x in the interval I, there exists a real number c between a and x such that

$$R_n(x)=\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

If there exists a real number M such that $\left|f^{(n+1)}(x)\right|\le M$ for all $x\in I$, then

$$|R_n(x)|\le\frac{M}{(n+1)!}|x-a|^{n+1}$$

for all x in I.